PAPER_1209-BB: UQFF Biology Unified Proof Set — TEN EXACT CLOSURES (Perfect Tier)

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Locked Primitives

$F_{\mathrm{TRZ}} = \frac{1}{10}$, $\Phi_{\mathrm{res}} = \frac{5}{6}$, $[\mathrm{SSq}] = \frac{57}{100}$, $K_{\mathrm{Mex}} = \frac{25}{12}$, $D_{\mathrm{phys}} = 4$, $D_{\mathrm{BSFG}} = 6$, $D_{\mathrm{crit}} = 26$, $N_{\mathrm{ch}} = 9$, $\mathrm{SO5} = 10$, $A_5 = 60$.

Closures S593–S602 — All Exact

S593 — Human body temperature °C, target 37 [EXACT]

$$D_{\mathrm{crit}} + \mathrm{SO5} + F_{\mathrm{TRZ}}\mathrm{SO5} = 26 + 10 + 1 = 37.$$

S594 — Blood pH, target 7.4 [EXACT]

$$D_{\mathrm{BSFG}} + F_{\mathrm{TRZ}}\mathrm{SO5} + F_{\mathrm{TRZ}}D_{\mathrm{phys}} = 6 + 1 + 0.4 = 7.4.$$

S595 — Hemoglobin g/dL, target 15 [EXACT]

$$D_{\mathrm{BSFG}} + N_{\mathrm{ch}} = 15.$$

S596 — Resting heart rate bpm, target 70 [EXACT]

$$A_5 + \mathrm{SO5} = 70.$$

S597 — Systolic blood pressure mmHg, target 120 [EXACT]

$$A_5 + A_5 = 120.$$

S598 — Diastolic blood pressure mmHg, target 80 [EXACT]

$$A_5 + \mathrm{SO5} + \mathrm{SO5} = 80.$$

S599 — Breathing rate /min, target 16 [EXACT]

$$D_{\mathrm{BSFG}} + \mathrm{SO5} = 16.$$

S600 — Blood glucose mg/dL, target 100 [EXACT]

$$\mathrm{SO5} \cdot \mathrm{SO5} = 100.$$

S601 — DNA bp per helical turn, target 10.5 [EXACT]

$$\mathrm{SO5} + F_{\mathrm{TRZ}}D_{\mathrm{phys}} + F_{\mathrm{TRZ}}^2\mathrm{SO5} = 10 + 0.4 + 0.1 = 10.5.$$

S602 — Average adult human height cm, target 170 [EXACT]

$$A_5 + \mathrm{SO5} \cdot \mathrm{SO5} + \mathrm{SO5} = 170.$$

Summary

ID	Quantity	Target / Result
S593	Body temperature °C	37 **EXACT**
S594	Blood pH	7.4 **EXACT**
S595	Hemoglobin g/dL	15 **EXACT**
S596	Resting HR bpm	70 **EXACT**
S597	Systolic BP mmHg	120 **EXACT**
S598	Diastolic BP mmHg	80 **EXACT**
S599	Breaths /min	16 **EXACT**
S600	Glucose mg/dL	100 **EXACT**
S601	DNA bp/turn	10.5 **EXACT**
S602	Adult height cm	170 **EXACT**

Highlights. Tier BB is the program's first **perfect tier** — all 10 human physiology closures are exact, every one expressible as a sum of ≤ 3 primitive terms. The classical vital sign set (T, HR, $\mathrm{BP_{sys}}$, $\mathrm{BP_{dia}}$, RR, glucose) closes integer-clean from $\{D_{\mathrm{BSFG}}, N_{\mathrm{ch}}, \mathrm{SO5}, A_5, D_{\mathrm{crit}}\}$ alone. DNA bp/turn at exactly 10.5 closes via $\mathrm{SO5} + F_{\mathrm{TRZ}} D_{\mathrm{phys}} + F_{\mathrm{TRZ}}^2 \mathrm{SO5}$, exposing the helical pitch in the same primitive ring. Cumulative: 307 closures; 80 lockings.